

Facile In Situ Chemical Cross-Linking Gel Polymer Electrolyte, which Confines the Shuttle Effect with High Ionic Conductivity and Li-Ion Transference Number for Quasi-Solid-State Lithium–Sulfur Battery

Tongwei Zhang, Jun Zhang, Shan Yang, Yuan Li, Ran Dong, Jialiang Yuan, Yuxia Liu, Zhenguo Wu, Yang Song, Yanjun Zhong, Wei Xiang, Yanxiao Chen,* Benhe Zhong, and Xiaodong Guo*



Cite This: *ACS Appl. Mater. Interfaces* 2021, 13, 44497–44508



Read Online

ACCESS |



Metrics & More



Article Recommendations



Supporting Information

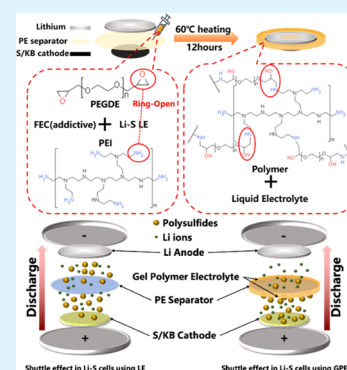
ABSTRACT: As a secondary Li-ion battery with high energy density, lithium–sulfur (Li–S) batteries possess high potential development prospects. One of the important ingredients to improve the safety and energy density in Li–S batteries is the solid-state electrolyte. However, the poor ionic conductivity largely limits its application for the commercial market. At present, the gel electrolyte prepared by combining the electrolyte or ionic liquid with the all-solid electrolyte is an effective method to solve the low ion conductivity of the solid electrolyte. We present a cross-linked gel polymer electrolyte with fluoroethylene carbonate (FEC) as a solid electrolyte interface (SEI) film formed for Li–S quasi-solid-state batteries, which can be simply synthesized without initiators. This gel polymer electrolyte with FEC as an additive (GPE@FEC) possesses high ionic conductivity (0.830×10^{-3} S/cm at 25 °C and 1.577×10^{-3} S/cm at 85 °C) and extremely high Li-ion transference number ($t_{\text{Li}^+} = 0.674$). In addition, the strong ability toward anchoring polysulfides resulting in the high electrochemical performance of Li–S batteries was confirmed in GPE@FEC by the diffusion experiment, X-ray photoelectron spectroscopy analysis (XPS), and scanning electron microscopy (SEM) mapping of the S element. Such a high ion conductivity (IC) gel polymer electrolyte enables a competitive specific capacity of 940 mAh/g at 0.2C and supreme cycling performance for 180 cycles at 0.5C, which is far beyond that of conventional poly(ethylene oxide)-based quasi-solid-state Li–S batteries.

KEYWORDS: *in situ* thermal cross-linking, gel polymer electrolyte, quasi-solid-state, lithium–sulfur battery, interfacial characteristics, high ionic conductivity

1. INTRODUCTION

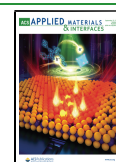
With increasing concerns regarding the issue of sustainable energy and a low-carbon society globally, the demand for replacing traditional gasoline vehicles with electric vehicles is gradually increasing. However, the requirement of a higher energy density becomes a major problem to solve in electric vehicles (EVs).^{1–4} Lithium is the most suitable material for anodes because of its high energy density and long cycle lifetime. On the other hand, for the cathodes, Li–S batteries are provided with unapproachable advantages such as the high natural abundance, high theoretical gravimetric energy density (2600 Wh/kg), and environmental benignancy of sulfur.^{5–8} Even though Li–S batteries possess the advantages mentioned above, the practical and commercial applications have been limited by many fundamental factors, including (i) the formation of Li dendrites caused by asymmetrical Li-ion deposition on the surface, which would not only decrease the cycle performance and Coulombic efficiency but also raise safety issues,^{9,10} and (ii) the shuttle effect from soluble polysulfides, which leads to a much smaller practical energy density than the theoretical energy density and inferior cycle ability.^{11,12}

Recently, to solve the problems mentioned above in common Li–S batteries, many strategies have been brought forward to inhibit polysulfide shuttle and safety issues caused by the Li dendrite. All-solid-state electrolytes, especially solid polymer electrolytes (SPEs), present as a burgeoning and convenient strategy to overcome the Li dendrite and safety issues.^{13–15} Nevertheless, key problems still remain when using the polymer solid electrolytes, including (i) rather low ionic conductivity polymer electrolytes at room temperature (RT) on account of the high crystallinity and slow chain mobility of the polymer matrix,^{16–18} (ii) unexpected side reactions between electrolyte/electrode interfaces,^{17,19,20} and (iii) the dissolution of polysulfides in the electrolyte system, leading to the uncontrollable shuttle effect.^{21,22}



Received: August 24, 2021

Published: September 10, 2021



One of the approaches to address these issues is to combine the polymer electrolyte with a certain amount of liquid electrolyte to achieve a quasi-solid-state electrolyte (QSSE) or a gel polymer electrolyte (GPE) in Li–S batteries. The QSSE or GPE shows a much higher ion conductivity than traditional polymer electrolytes and provides great surface contact of the lithium metal by improving the formation of the solid electrolyte interface (SEI).^{9,23,24} Some common polymers, such as poly(ethylene oxide) (PEO),²⁵ poly(acrylonitrile) (PAN),²⁶ and poly(vinylidene fluoride hexafluoropropylene) (PVDF-HFP),^{27,28} could be applied to QSSE as the matrix. PEO polymer is a conventional selection as host materials for building QSSE because of the strong ion–dipolar interactions between the ether chain $\text{CH}_2\text{CH}_2\text{O}$ unit and lithium ions, which provide paths for lithium-ion transport. Unfortunately, the ether chain could not inhibit the shuttle effect of polysulfides.^{21,29}

In this article, we introduce a new cross-linked gel polymer electrolyte with supreme high ionic conductivity and the ability to inhibit the shuttle effect. As the previous article reported, the inhibition of the shuttle effect comes from the binding ability between polysulfides and the functional groups of the gel polymer electrolyte.^{30–32} The binding energies of pyridinic nitrogen (–pN), pyrrolic nitrogen (–rN), ketone (–kO), carboxylic (–aO), and hydroxyl (–hO) are much higher than those of the other nitrogen and oxygen groups in nitrogen-doped carbon materials, and we know that the functional groups with N and O can play the same role as the nitrogen or oxygen atoms doped into carbon materials.^{33,34} The interactions between these high-electronegativity atoms of functional groups and Li^+ are well known as the Li–bond. These atoms with an extra pair of electrons act perfectly as the Lewis-base site. Therefore, they can interact with the strong Lewis acid, which is the terminal Li atom of lithium polysulfide.^{35,36} As a result, we used polyethyleneimine, branched (PEI, branched), which consists of amine groups and poly(ethylene glycol)diglycidyl ether (PEGDE) with epoxy groups as cross-linked agents. This gel polymer electrolyte is prepared through a ring-opening addition polymerization of amine groups and epoxy groups, which creates the ether chain and the amino group that have intensive interactions with Li ions to offer paths for Li-ion transport, and lone pair electrons of the N atom from amino groups facilitates ion transport by the Li–bond mentioned before.³⁷ Apart from amino groups, there are –NH–, C–N–C, –OH, and C–O–C in the polymer matrix, which provide the Lewis-base sites to restrict polysulfides within the cathode side.³⁸ The suppression of the polysulfide shuttle effect is illustrated in Figure 1. Due to the above-mentioned aspects of this gel polymer electrolyte, the cycle

performance of a quasi-solid-state Li–S battery was realized with 80% capacity retention after 180 cycles at 0.5C, which is far beyond that of the pristine liquid electrolyte. Otherwise, the ionic conductivity of this gel polymer electrolyte is higher than that of the pristine liquid electrolyte at room temperature (RT), which shows a supreme advantage compared with the traditional solid-state electrolyte.³⁹ Furthermore, the strong absorptivity toward polysulfides of this gel electrolyte was verified through the visible adsorption of 20 mM Li_2S_8 .

2. EXPERIMENTAL SECTION

2.1. Materials. Polyethyleneimine (Aldrich Chemical Co., $M_w = 1000$, branched) and poly(ethylene glycol)diglycidyl ether (Aldrich Chemical Co., $M_w = 500$) were kept in vacuum drying for 4 h at 90 and 50 °C, respectively. A commercial liquid electrolyte with 1.0 M LiTFSI in 1,2-dimethoxyethane/1,3-dioxolane (DME/DOL) (1:1 by volume, battery grade) and 0.1 M LiNO_3 was purchased from DUODUO chemical Co. and was used without further treatment. The electrolyte additive fluoroethylene carbonate (DUODUO chemical Co, battery grade) was stored in a glovebox.

2.2. Synthesis of the Gel Polymer Electrolyte Precursor. The PEGDE and PEI were dropped into a small bottle with a weight ratio of 1:1. The obtained mixture of these two monomers was added into the liquid electrolyte at a weight concentration of 10 wt %. Then, extra fluoroethylene carbonate (FEC) at a concentration of 4 wt % was added into the abovementioned mixture, followed by stirring for 12 h. Finally, the precursor solution was stored at an ambient temperature to use in the cell assembly procedure.

2.3. Preparation of the S/KB Composite. The cathode active material was prepared by the melt diffusion method in a tube furnace. Sublimed sulfur and Ketjen Black (KB EC-600JD) were added with a weight ratio of 7:3 and then treated by ball-milling for 1 h. This mixture was then sealed into a tube furnace at a temperature of 155 °C for 12 h with the flow of Ar gas.

2.4. Preparation of the S/KB Cathode. The cathode slurry of this Li–S battery was synthesized by adding the S/KB composite, conductive agent (Acetylene Black), and binder (poly(vinylidene fluoride), PVDF) at a weight ratio of 70:20:10, while the *N*-methyl pyrrolidone (NMP) was selected as the dispersant. The prepared slurry was ball-milled for 40 min and coated on an aluminum foil using the blade-casting method. Then, this aluminum foil coated by the slurry of the cathode was dried in a vacuum oven at 60 °C for 12 h. The sulfur cathode was taken out and then cut into a disk with a diameter of 13 mm, while the S loading weight of each cathode disk was around 1.0 mg/cm^2 .

2.5. Fabrication of the Quasi-Solid Li–S Battery. The sulfur cathode, PE separator, and metallic lithium anode were assembled in cells (CR2025) from the bottom to the top in a glovebox, which provided an atmosphere of Ar where the H_2O and O_2 contents were less than 0.5 ppm. Fifty microliters of the abovementioned gel polymer electrolyte precursor was added into the cells during the assembly procedure. The cells were kept still for a few hours to let the electrolyte precursor fully infiltrate within the PE separator, and then they were heated at 60 °C for 24 h in a vacuum oven to induce polymerization. The other matched cells using commercial liquid electrolytes (1.0 M LiTFSI in DME and DOL with a volume ratio of 1:1) to replace the gel polymer electrolyte precursors were carried out through the same method mentioned above.

2.6. Materials Characterization. Scanning electron microscopy (SEM, Hitachi, S-4800) was used to measure the separator's morphology and lithium surfaces, and energy-dispersive spectroscopy (EDS) was coordinated with SEM images to distinguish the distribution of the S element. The surface composition was determined by X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi, Al $K\alpha$). Functional groups and chemical bond vibration were verified by Fourier transform infrared (FTIR) and Raman spectroscopies (FTIR: Thermo Fisher Nicolet Is 5, Raman: Thermo Fisher Dxi). Nuclear magnetic resonance (NMR) spectra of monomers and

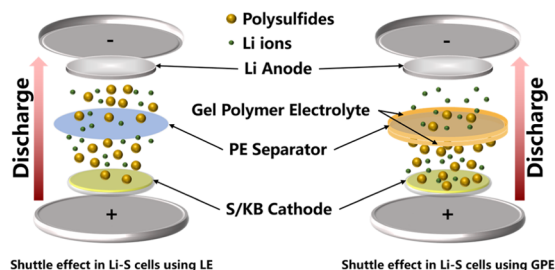


Figure 1. Illustration of the shuttle effect in Li–S cells using LE and GPE.

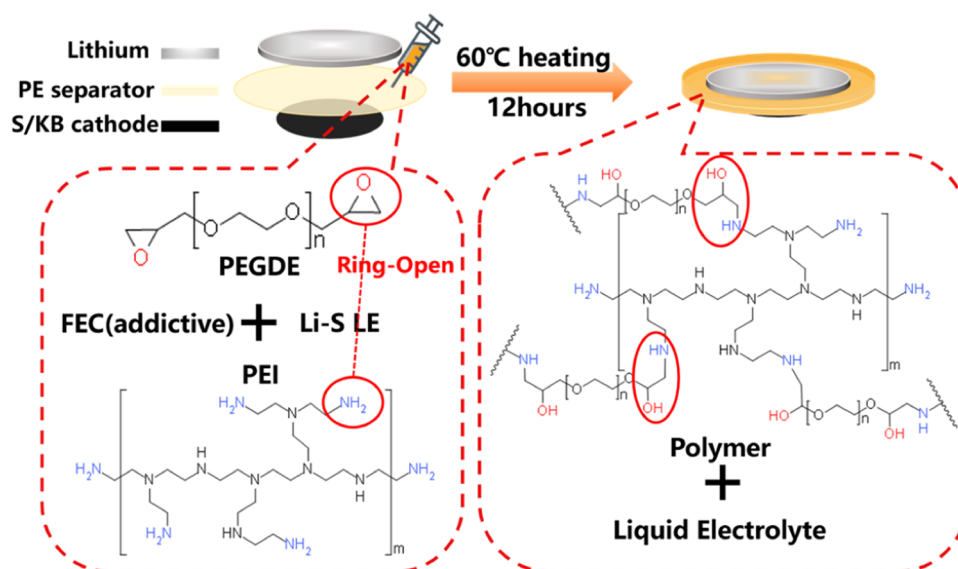


Figure 2. Synthesis scheme of the PEI and PEGDE by thermal in situ polymerization and the illustration scheme of the gelation experiment.

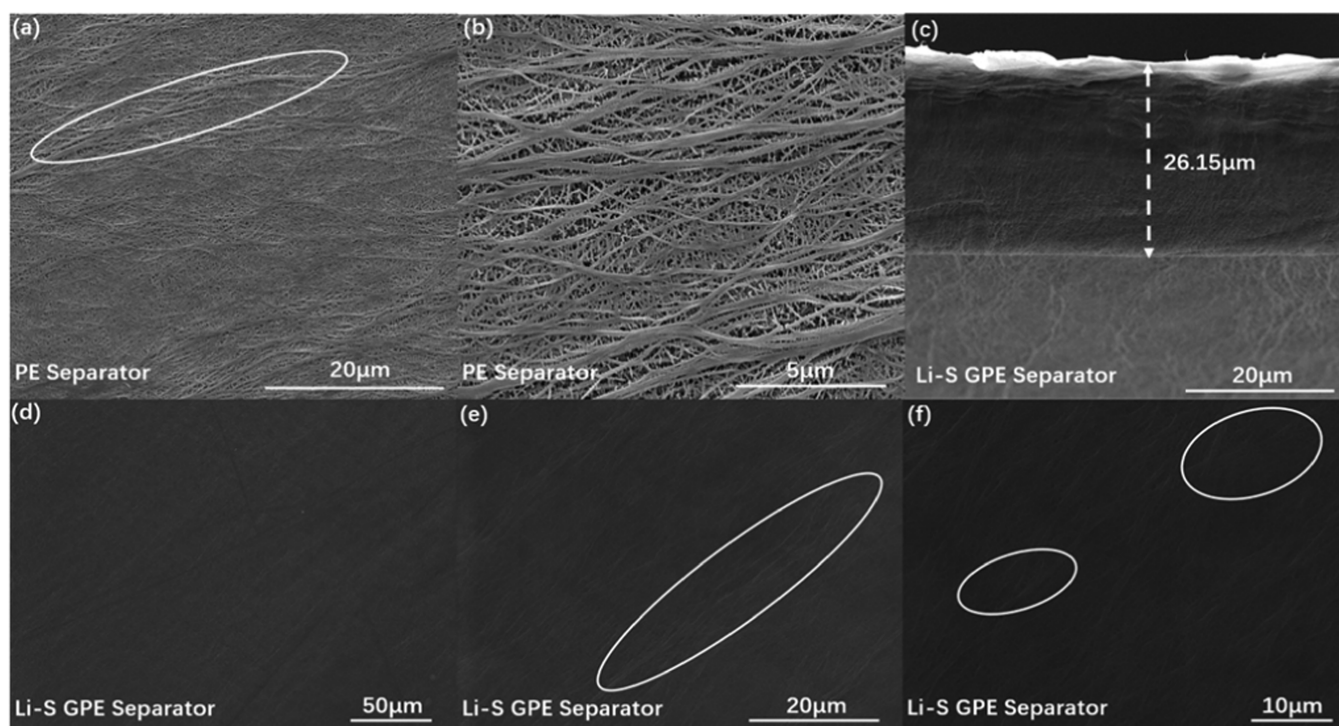


Figure 3. SEM images of (a, b) surface on the PE separator, (c) thickness of the GPE membrane, and (d–f) surface on the GPE membrane.

gel polymer were measured by a Bruker AVANCE AV 400 MHz spectrometer. The thermal stability of the gel polymer electrolyte was testified by a thermogravimetric analyzer (TGA, METTLER 1100LF).

2.7. Electrochemical Measurements. The charge/discharge measurement of the batteries was implemented with the battery measurement system (NEWARE). The cyclic voltammetry (CV) was recorded with an electrochemical workstation (CS2350H Bipotentiostat, Corrtest) from 1.7 to 2.8 V at a scanning rate of 0.1 mV/s or 0.1 to 0.5 mV/s. Electrochemical impedance spectroscopy (EIS), whose sweeping frequency was from 1 mHz to 10 MHz and an AC amplitude of 5 mV at open circuit potential, was implemented with a Zahner Autolab electrochemical workstation. The electrochemical stability of GPE performed with the SS|GPE||Li coin cell at a scanning rate of 10 mV/s was measured by linear sweep voltammetry (LSV).

The Li-ion transference number (t_{Li^+}) of GPE and LE was calculated according to the equation⁴⁰

$$t_{Li^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} \quad (1)$$

where I_0 and R_0 are the initial current and initial interfacial resistance, respectively, ΔV is the applied polarization voltage (set as 10 mV), and the I_{ss} and R_{ss} are the steady-state current and steady-state interfacial resistance of the cell. The ionic conductivities of the GPE and LE were analyzed by an AC impedance of ion blocked cells at temperatures ranging from 25 to 85 °C, with an interval of 15 °C, and were calculated according to the equation⁴¹

$$\sigma = \frac{L}{RS} \quad (2)$$

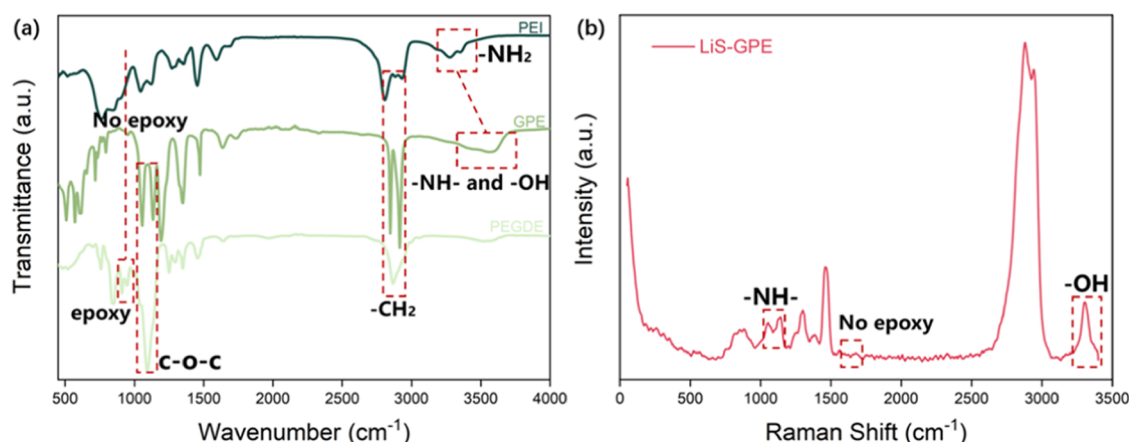


Figure 4. (a) FTIR spectra of PEI, PEGDE, and Li-S GPE. (b) Raman spectrum of Li-S GPE.

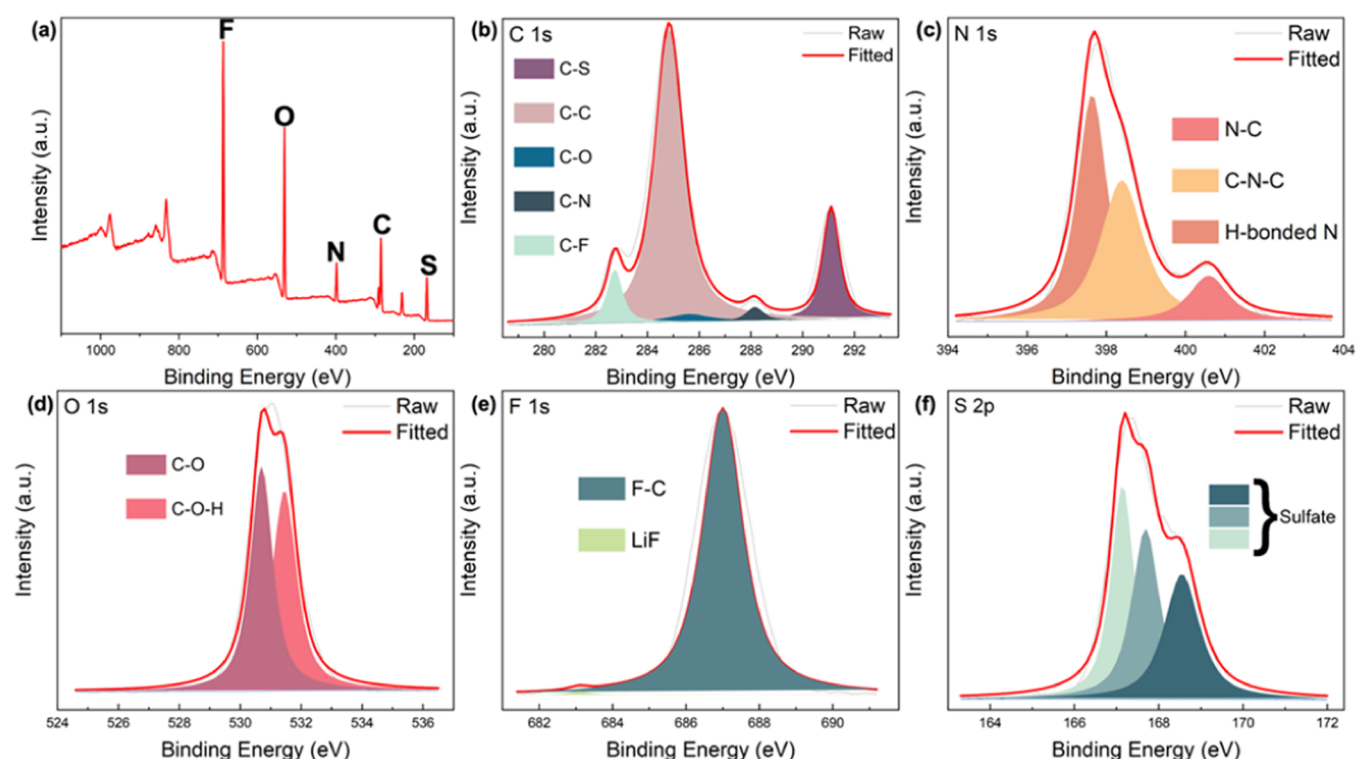


Figure 5. Refined XPS results of different elements in the GPE membrane: (a) whole spectra, (b) C 1s, (c) N 1s, (d) O 1s, (e) F 1s, and (f) S 2p.

where σ is the ionic conductivity of the electrolyte, L is the thickness of the electrolyte or the distance between two electrodes, R is the intercept at the real axis in the AC impedance, and S is the surface area of the electrolyte.

3. RESULTS AND DISCUSSION

3.1. Polymerization. The schematic illustration and the polymerization process are shown in Figure 2. The gel electrolyte precursor consisted of PEI, PEGDE, FEC, and liquid electrolytes. The GPE molecular structure with polar nitrogen functional groups and ether chains could promote the Li⁺ transport as they could coordinate with lithium ions. The gel formation was visually confirmed (Figure S1) as the shallow yellow precursor at the bottom of the glass bottle that turned to brown gel that would not fall or collapse after the thermal gelation.

The SEM images of the in situ GPE membrane and PE separator are shown in Figure 3. The surface of the PE separator is full of fibrous holes. On the other hand, there are no holes and cracks on the surface of the GPE membrane, which means the holes in the PE separator are all filled by the in-situ gel polymer. The SEM images showing the thickness of the PE separator and the GPE membrane are shown in Figures 3c and S2. The thickness of the PE separator is 12.21 μm , and the thickness of the GPE membrane increases to 26.15 μm . The increase in the thickness of the GPE membrane can be concluded as the thickness of the gel polymer. Meanwhile, the thickness of GPE is very smooth, indicating a uniform coating of the gel polymer on the PE separator.

3.2. Structure Analysis and Suppression on the Shuttle Effect. FTIR spectroscopy of monomers and gel polymer was performed to elucidate the chemical cross-linking polymerization and the structural changes among PEI and

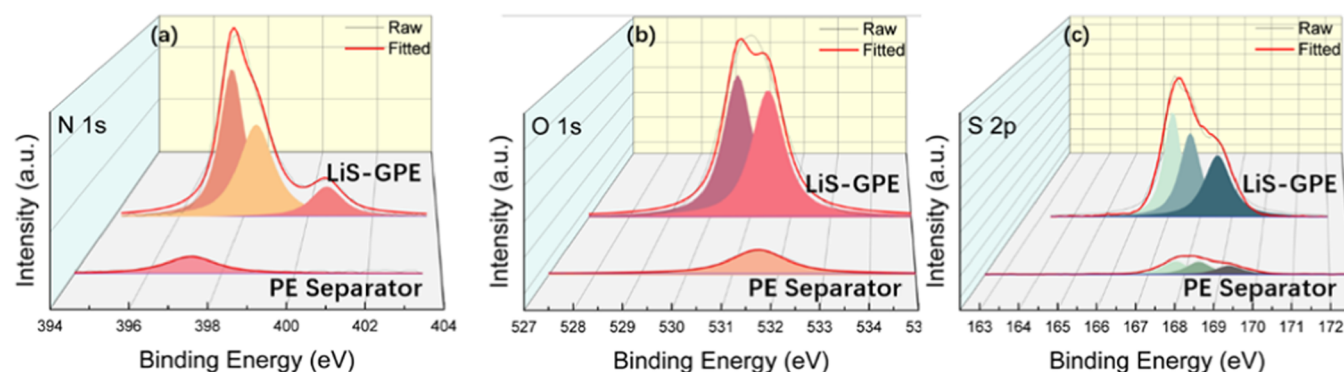


Figure 6. XPS intensity differences of Li-S GPE and PE separator on (a) N, (b) O, and (c) S elements.

PEGDE; the FTIR spectra are shown in Figure 4a. The peaks at 1150 cm^{-1} among PEGDE and GPE were assigned to the typical C–O–C stretching vibration in the ether chain PEG unit, and the peaks near 1500 cm^{-1} were assigned to the bending vibration of —NH_2 .⁴² In GPE, a broad peak between 3280 and 3700 cm^{-1} was verified as the stretching vibration of O–H and N–H.⁵ After the addition polymerization, the characteristic peaks corresponding to the epoxy groups of PEGDE at 907 and 850 cm^{-1} disappeared in the spectrum of GPE,⁴³ which indicates the successful thermal reaction between the epoxy groups in PEGDE and the amino groups in PEI. Therefore, this GPE was triumphantly obtained within the cell after the chemical cross-linking reaction. Furthermore, Figure 4b is the Raman spectrum of the PEGDE-PEI lithium–sulfur gel solid electrolyte membrane. The result of the Raman spectrum was consistent with that of the FTIR spectrum. In the Raman spectrum, the Raman shift of the epoxy group should be nearly located at 1610 cm^{-1} , but there is no absorption peak at 1610 cm^{-1} in this Raman spectrum of the Li–S gel polymer.⁴⁴ The midstrength peak at the Raman shift of 3303 cm^{-1} and the peak at 1141 cm^{-1} , respectively, represent the stretching vibration peak of the hydroxyl group (—OH) and the stretching vibration peak of the secondary amine (—NH—). Bulk PEGDE and multibranched PEI have been successfully thermally polymerized to form a macromolecular polymer network.

In addition, the types and quantity of H atoms in different environments of monomer PEGDE, branched PEI, and this gel polymer are determined by nuclear magnetic hydrogen spectroscopy (NMR), as shown in Figure S3. In the hydrogen NMR spectrum of the monomer PEGDE, there is the highest peak at the chemical shift of 3.52 ppm , which is verified as the main carbon chain skeleton (C–O–C groups) of PEGDE. The two peaks at 2.54 and 2.73 ppm are verified as the two H atoms in epoxy. At the same time, in the hydrogen NMR spectrum of branched PEI, the broad and short peak at 1.6 ppm corresponds to the peaks of the two hydrogen atoms in the primary amine (R—NH_2). However, these H atoms of two major functional groups that participate in the ring-opening addition reaction to form our gel polymer electrolyte cannot be found in the hydrogen solid-NMR spectrum of this GPE, but there is the same main peak at 3.74 ppm in GPE, indicating that the C–O–C skeleton was inherited in our GPE membrane. Meanwhile, there is a small peak located at 6.02 ppm , which is attributed to —OH groups. As shown in the sketch of ring-opening polymerization, the appearance of —OH is evident to prove the formation of GPE.

These results are further confirmed by the X-ray photoelectron spectroscopy (XPS), as presented in Figure 5. The surface of the gel electrolyte within the cell after 180 cycles was analyzed by XPS measurement. In the full survey spectrum of XPS, distinct peaks of C 1s, N 1s, O 1s, F 1s, and S 2p were observed. The deconvolution of the N 1s signal can be well resolved into three peaks, which can be identified at 397.7 , 398.3 , and 400.6 eV , corresponding to the single bond between nitrogen and carbon (N–C), the N bonded by two C (C–N–C) in the cross-linked gel polymer, the H-bonded N of the amino groups, respectively.^{45–47} Therefore, the absorption of polysulfides can be greatly improved as the existence of both N–C and C–N–C due to their high binding energies of the N atom. In the meantime, the presence of the C–O–C (PEG units) peak at 531.5 eV and the C–OH peak at 531.7 eV is verified by the O 1s XPS spectra as well,⁴⁸ which also confirm the addition polymerization (the amino groups in PEI and the epoxy groups in PEGDE) within cells. Meanwhile, the peak at 687.8 eV for F 1s can be attributed to the formation of fluorinated carbamate. After that, several peaks are detected between 168.5 and 170.3 eV , indicating that polysulfides are anchored in this gel polymer electrolyte.^{17,45}

To compare the element content, the XPS images of the same elements in both the Li–S GPE membrane and the PE separator in the Li–S LE battery on the same axis were compared in terms of strength (Figure 6). It can be found that the peak intensity obtained on the GPE membrane of the three elements compared is much higher than that in the PE separator. The content of nitrogen, oxygen, fluorine, and other elements on the gel electrolyte membrane is much higher than that in the PE separator, and the structure of nitrogen and oxygen atoms is the same as the specific functional group mentioned above, indicating that the gel polymer is rich in polysulfides. The material produces an electrostatic effect to adsorb the functional groups of polysulfides and successfully confine polysulfides on the electrolyte membrane.^{49,50} On the other hand, PE separators do not have these functional groups, as shown in the XPS results in Figure S4. There are more S elements on the GPE membrane of sulfur batteries. We compared the S element at the anode side of the separator and the gel electrolyte membrane after the cycle in Figure S5. We could find that there was more S element on the contact surface of the separator and the anode, which further proves that the PEGDE-PEI gel electrolyte membrane can inhibit the shuttle effect of polysulfides, and theoretically, it can promote the electrochemical reaction in lithium–sulfur batteries and improve the Coulombic efficiency and cycle performance of gel lithium–sulfur batteries.

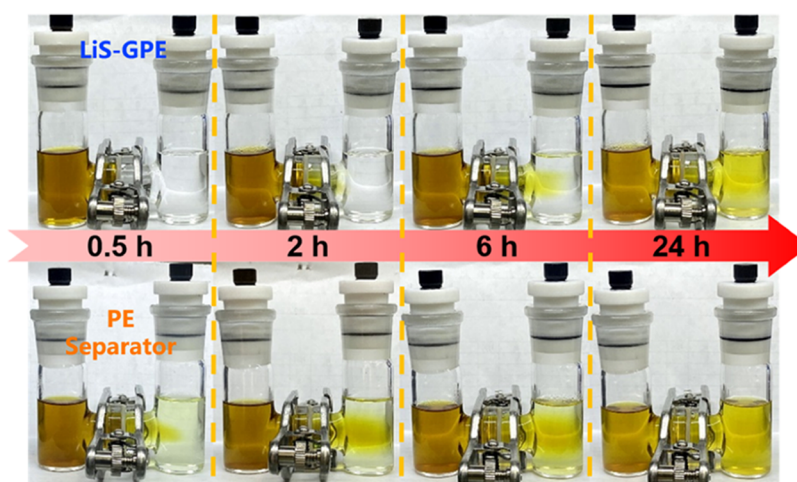


Figure 7. Diffusion experiment images of the GPE@FEC membrane and the PE separator for 24 h.

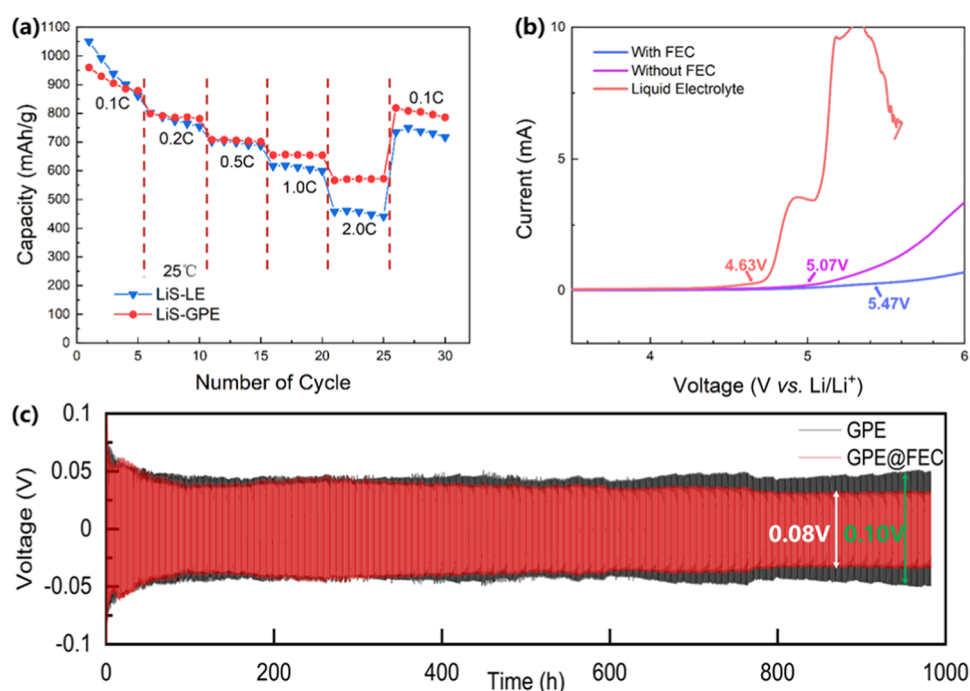


Figure 8. (a) Rate performance of batteries assembled with GPE, GPE@FEC, and LE; (b) electrochemical stability windows of GPE, GPE@FEC, and LE at a scanning rate of 1 mV/s; (c) chronopotentiometry results for Li/GPE@FEC/Li and Li/GPE/Li symmetrical cells at 25 °C under a current density: 0.1 mA/cm².

The strong immobilization of polysulfides by the GPE can be verified through the diffusion experiment carried out in an H-type glass cell. Fifteen milliliters of Li₂S₆ solution was added into the left tube of the communicating vessel, and an equal volume of DOL and DME mixture was added into the right tube as the blank electrolyte. A gel polymer electrolyte membrane and a commercial PE separator were clamped by the two tubes on the left and right and then left to stand for 24 h in a glovebox to exclude the influence of water and oxygen. As shown in Figure 7, conspicuous color changes of the blank electrolyte in the right tube were observed. After half an hour, a small amount of polysulfides diffused through the PE separator, while the blank tube of the GPE membrane had no apparent color change. After 6 h, the color of the PE separator turned pale yellow. In contrast, only a few polysulfides passed through the GPE membrane. Within 24

h, it could be observed that the yellow part of the PE separator was much deeper than that of the GPE membrane, indicating that polysulfides were effectively absorbed and could merely pass through the GPE membrane. With these phenomena and the result of XPS N 1s peaks, the strong absorption toward polysulfides of the GPE membrane could be certified.

To prove the adsorption effect of the PEGDE-PEI gel electrolyte membrane on polysulfides (PSs) from the perspective of the S element content, we disassembled the Li-S GPE battery and the ordinary Li-S LE battery and took them out separately. For the GPE membrane and the ordinary PE separator, the surface of the two membranes in contact with the Li-S cathode material was viewed by SEM, and the mapping diagram of the element S was taken to distinguish the content and distribution of the element S on the surface of the two membranes. It can be seen from Figure S6 that the S

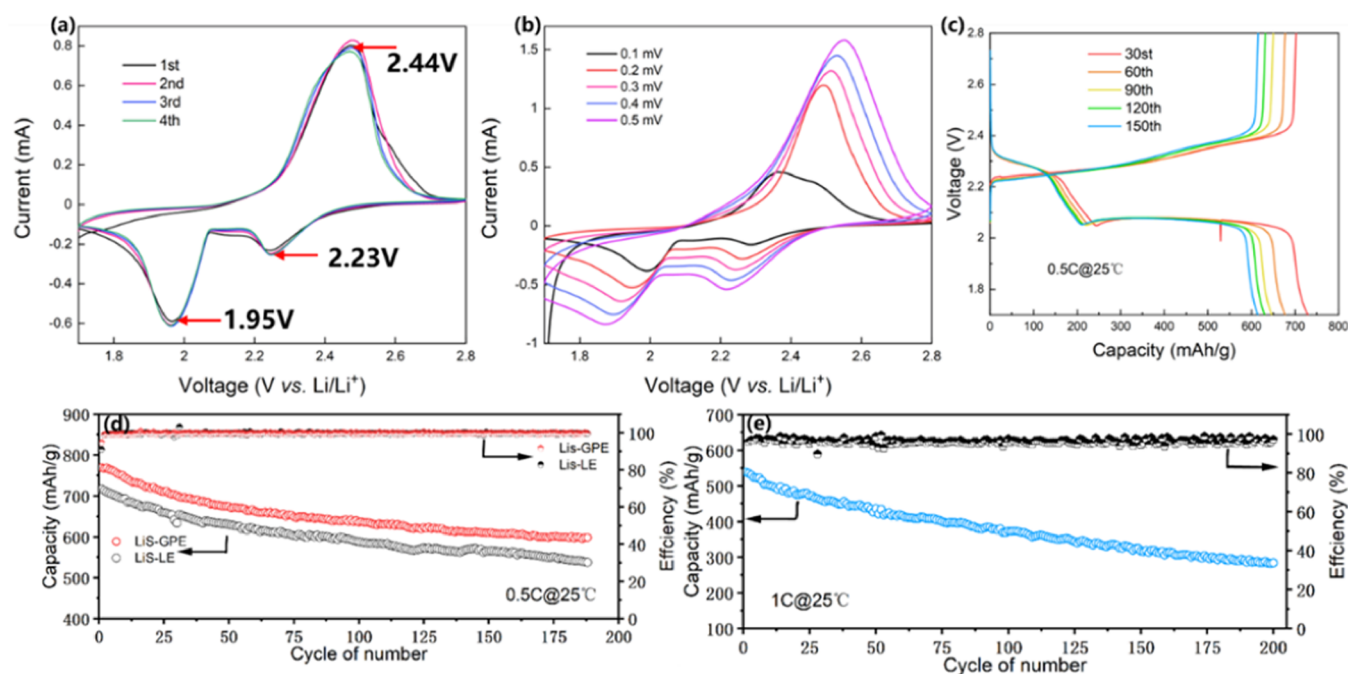


Figure 9. CV curves of S//Li batteries under a voltage range of 1.7–2.8 V (vs Li/Li⁺): (a) scanning rate: 0.1 mV/s for four cycles; (b) scanning rate: from 0.1 to 0.5 mV/s; (c) discharge profiles of S//Li monopolar batteries at a rate of 0.5C for different cycles; (d) cycling performance at 0.5C at 25 °C; and (e) cycling performance at 1.0C at 25 °C.

element distribution on the surface of the gel electrolyte membrane is very dense, while the S element distribution on the surface of the PE separator is relatively sparse. It proves that in the same cyclic discharge stage, this PEGDE-PEI GPE can play a significantly better limitation on the shuttle effect of PSs participated in the electrochemical reaction so that the polysulfides are concentrated in a large amount between the electrolyte membrane and the positive pole piece instead of being arbitrary. The polysulfide shuttles to the metal lithium negative electrode under the action of the electric field force and the concentration gradient, so the positive side surface of the gel electrolyte membrane has significantly more S element distribution than the ordinary separator.

3.3. Thermal Stability. The thermal stability of the solid polymer electrolyte is quite a critical character because of the safety problems of commercial lithium batteries. The thermal stability of this gel polymer electrolyte membrane was measured by TGA analysis in a N₂ atmosphere, and the temperature is set from 25 to 600 °C at a heating rate of 10 °C/min. With the derivative thermogravimetric (DTG) curve, the decomposition temperature of this membrane can be observed from Figure S7. The membrane started decomposing at 346.2 °C, which means this gel polymer electrolyte is thermally stable up to about 350 °C. Apparently, this gel polymer electrolyte membrane can be fully competent for application under high-temperature conditions compared with the commercial liquid electrolyte.

3.4. Electrochemical Performance. To investigate the electrochemical performance of this gel polymer electrolyte, a half-cell of sulfur/GPE/Li was assembled in the following experiment. Figure 8a shows the cells' performances under diverse current densities (0.1C to 2C) with the voltage confined from 1.7 to 2.8 V based on different electrolytes (Li–S GPE and LE). The discharge capacity delivered by the cells assembled by the sulfur/GPE with FEC/Li was more superior to that of cells using common LE among all different rates.

Therefore, on these rate performances, the equal ratio of PEI and PEGDE can ensure enough networks to restrain the liquid electrolyte but apply flexibility for the whole GPE structure, which benefits the application of this kind GPE cells.

In addition, the electrochemical stability of this in-situ-polymerized gel electrolyte for practical usage with high-voltage cathodes is another important parameter that could be investigated by linear sweep voltammetry (LSV) measurements. A SS/GPE/Li battery was assembled and tested by LSV from 1 to 6 V (vs Li/Li⁺) under a scan rate of 0.1 mV/s. The obtained results for GPE with FEC, GPE without FEC, and common liquid electrolyte are shown in Figure 8b and reveal that GPE with an equal weight of PEGDE and PEI as well as the addition of FEC displayed excellent electrolyte chemical stability without any pronounced anodic peak until 5.5 V (vs Li/Li⁺). In contrast, the same method for the liquid electrolyte (LE) showed that oxidation already occurred at 4.63 V. Moreover, the anodic peak of the GPE without FEC occurred at 5.07 V, which means that the lack of FEC leads to the poorer stability of the electrode interface.

To evaluate the suppression effect of the in situ gel polymer electrolyte with/without FEC on Li dendrite growth, the lithium stripping and plating behavior was testified by the polarization of the Li/Li symmetrical test.⁵¹ These Li/GPE/Li batteries were conversely charged and discharged for 0.5 h periods with a constant current density of 0.1 mA/cm² at 25 °C, and the results are shown in Figure 8c. Each battery with/without FEC exhibits extreme long-term stability beyond 1000 h, which displayed a rather flat stripping/plating curve. It can be concluded from these results that both GPE with or without FEC as an additive showed great durability and electrochemical stability toward lithium. The only difference between the two curves is the polarization voltage. The GPE with the addition of FEC showed a lower polarization voltage (about 0.03 V) than GPE without the addition of FEC (over 0.04 V). As is well known that the increase of polarization reflects the

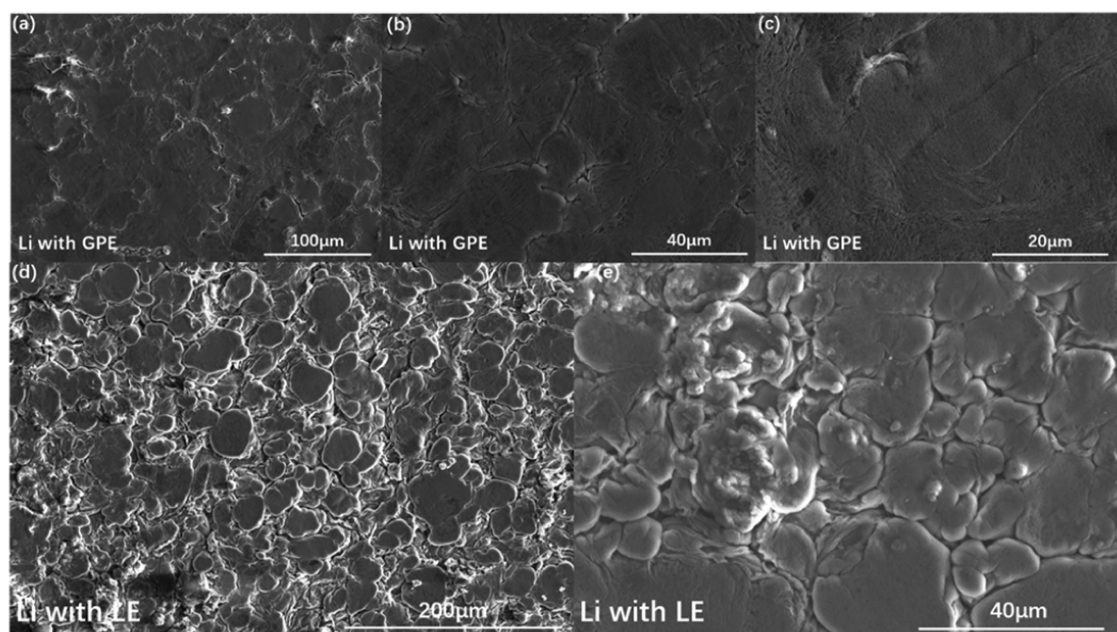


Figure 10. SEM images of the lithium surfaces assembled with different electrolytes at 0.5C after 200 cycles: with GPE@FEC (a–c) and with LE (d, e).

increase of the battery internal resistance, the addition of FEC restrained the growth of internal resistance by inducing the formation of a SEI between electrodes.

Cyclic voltammetry (CV) measurement of half-cell sulfur/GPE/Li using GPE with FEC carried out at 25 °C is shown in Figure 9a,b, in which the potential ranges between 1.7 and 2.8 V, while the scan rate was set as 0.1 mV/s for four cycles and the different scan rates were set as 0.1–0.5 mV/s for a single cycle. The CV curves illustrate the electrochemical process for sulfur, and there was no other excessive oxidation or reduction from the CV curves of this gel polymer cell except an oxidation peak at 2.44 V, while there are two reduction peaks at 2.23 and 1.95 V. During the discharging process, the cathodic peak at 2.23 V was associated with the reduction of S_8 to polysulfide (PS) intermediates, and the peak at 1.95 V was corresponding to the reduction of Li_2S . The oxidation peak at 2.44 V could be attributed to the transformation of short-chain to long-chain PSs. In addition, the CV curves at the same scan rate (0.1 mV/s) overlapped very evenly, demonstrating the highly reversible behavior and electrochemical stability of the GEP membrane with FEC. In the meantime, the CV curves at different scan rates (0.1–0.5 mV/s) showed a slight shift in all of the oxidation and reduction peaks, indicating that the reactions were influenced as the scan rate became larger. However, with no other excessive peak appearing, the electrochemical stability at a high rate of GPE cells could be confirmed.

The cycling performance of this gel polymer battery assembled with GPE with FEC was evaluated in the following experiment. At first, the assembled cells were treated under a preconditioning cycle at 0.05C to activate the Li ions in Li–S cells between the same voltage range mentioned above. After two cycles of activation at 0.05C, the batteries were charged gradually up to 0.5C over the same voltage range. Figure 9c shows a typical charge/discharge platform with a voltage profile of the 30th, 60th, 90th, 120th, and 150th charge/discharge cycles of the GPE Li–S batteries. The battery delivered a discharge capacity of over 720 mAh/g at the 30th cycle. While the discharge capacity faded to about 630 mAh/g

at the 150th cycle. All five curves of each cycle can nearly match each other, especially the voltage platform, which means there was no apparent voltage decadence during the long-term cycles. This result fits with the CV curves according to highly reversible behavior and electrochemical stability of the GPE cells. Figure 9d shows the cycling performance in the batteries prepared with the GPE and FEC. Distinctly, the capacity retention of GPE with FEC cells could be calculated as 79.1% after 180 cycles, and this retention rate was slightly improved compared to other lithium–sulfur gel batteries.^{52–54} In addition to gel lithium–sulfur batteries, this result is close to the performance of all-solid-state lithium–sulfur batteries that have achieved better performance.^{55,56} Meanwhile, the Coulombic efficiency was maintained at over 99.5% throughout cycling and stabilized with the cycle number. On this condition, the strong absorption of polysulfides (PSs) by specific functional groups like $-NH_2$, $-NH-$, $-OH$, $C-O-C$, etc., which possess high binding energy with PSs, could be a vital factor for the supreme capacity retention and high Coulombic efficiency.

In Figure 10, the SEM images on surface morphologies of the lithium in the batteries using GPE@FEC and LE were obtained. Obviously, the surface of lithium of GPE@FEC batteries is much smoother than that of LE, indicating that the growth and accumulation of Li dendrites are well confined due to the usage of GPE@FEC. On the contrary, there are much more uneven dendrites and cracks in the surface of LE batteries. In Figure S8, we could find that the cross-section of the Li anode in the gel lithium–sulfur battery and common lithium–sulfur battery has a significant difference in flatness after cycling. It can be proved that this PEGDE-PEI gel electrolyte membrane can indeed inhibit the formation of lithium dendrites during the cycle and improve the cycle performance and safety of the lithium–sulfur electrolyte. The reason is that the PEGDE-PEI gel polymer has a higher Li^+ transference number (t_{Li^+}). At the same time, due to the existence of N- and O-atom-containing functional groups, it can provide more lithium-ion transport channels, and the

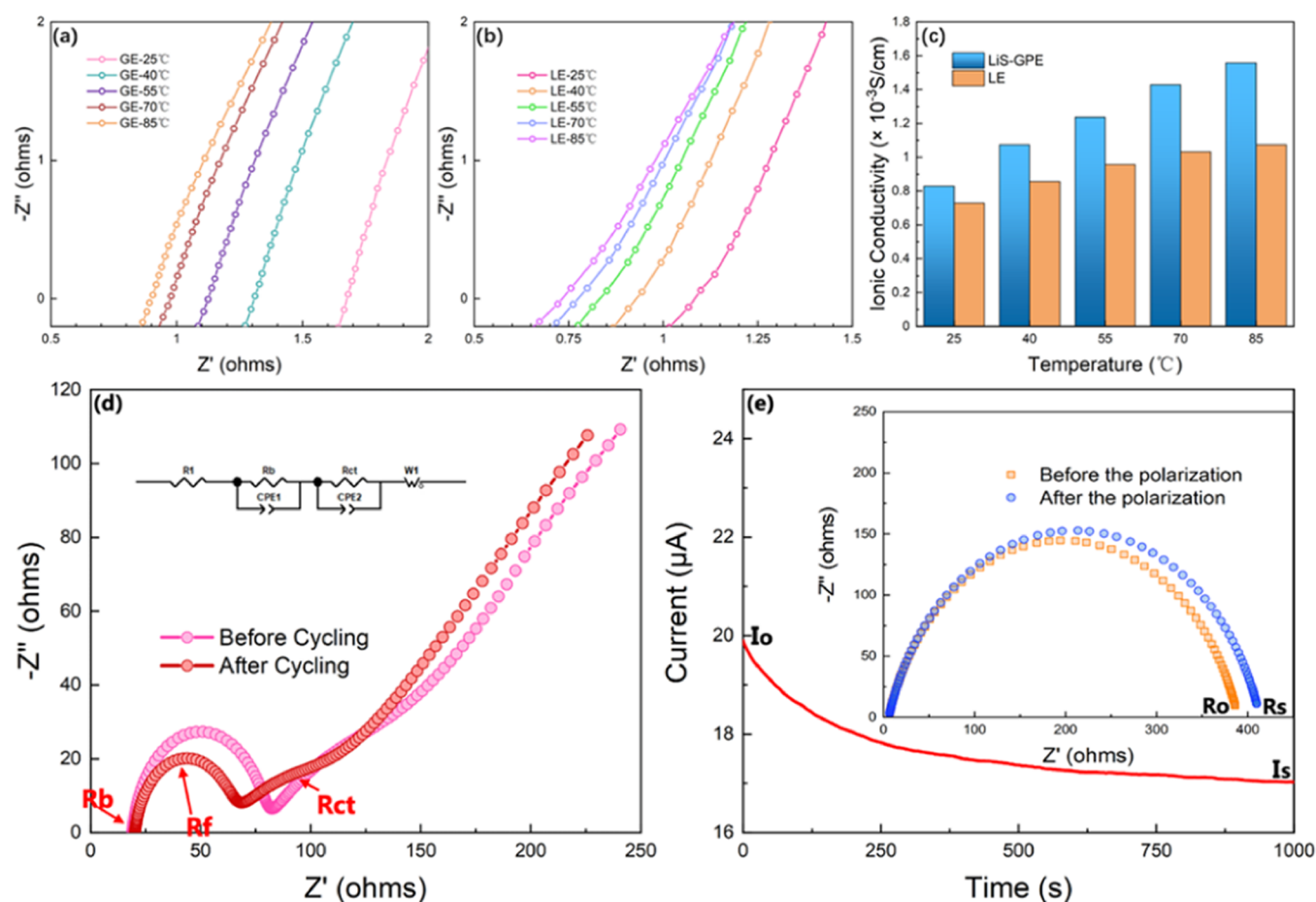


Figure 11. Electrochemical impedance spectra: (a) EIS of SS/GPE@FEC/SS from 25 to 85 °C; (b) EIS of SS/LE/SS from 25 to 85 °C; (c) the histogram about ionic conductivity of the GPE@FEC membrane and LE; (d) EIS of S batteries assembled with GPE@FEC before cycling and after 200th cycles; (e) chronoamperometry of a Li/GPE@FEC/Li battery with potential steps of 10 mV at 25 °C. Inset: the AC impedance spectra before and after polarization.

contact between the multibranched polymer structure and the metal lithium interface is closer to optimize the lithium-ion transfer site on metallic lithium.

To investigate the resistance of GPE batteries, the AC impedance spectra were obtained at the charge state after the activation procedure and after 200 cycles. The two AC impedance curves consist of two matched semicircles, which could be attributed to the resistance of Li^+ ions moving through the SEI film (R_s) at the lithium anode surface and a straight line, which referred to charge transfer resistance at the electrode–electrolyte interface (R_{ct}). Meanwhile, the electrolyte bulk resistances (R_b) were estimated as the intercept of the real axis under the high-frequency range. As shown in Figure 11d, the electrolyte resistances (R_b) at the initial cycle (with a value of 19.04 Ω) and after 200 cycles (with a value of 19.88 Ω) indicated that this gel polymer electrolyte did not undergo structure collapse and leakage of the liquid electrolyte after long-term cycling. In addition, this electrolyte resistance value is low enough as a quasi-solid polymer electrolyte to improve the ionic conductivity and rate performance. However, the resistance of the SEI film (R_s) after 200 cycles (with a value of 43.29 Ω) was obviously smaller than that at the initial cycle (with a value of 59.82 Ω). The resistances of charge transference at the electrode–electrolyte interface (60.75 Ω at the initial cycle while 41.97 Ω after 200 cycles) demonstrated the same tendency that resistances might

become lower with the increase of the charge–discharge cycle. These results demonstrated that our in situ cross-linking GPE was very effective in maintaining good interfacial contact between two electrodes and GPE. Additionally, the decreasing resistance of R_b and R_{ct} may result in the presentation of FEC by facilitating the formation of a smooth SEI, which benefits Li^+ movement through the anode surface and interface. As a result, the battery with GPE@FEC exhibited superior cycling performance compared with traditional gel electrolyte-based batteries.

The variability between ionic conductivity (IC) and the temperature was testified by symmetric block stainless steel (SS) electrodes from 25 to 85 °C, and the AC impedance results are shown in Figure 11a,b.^{57,58} We can find that the ionic conductivity of GPE@FEC and LE reveal temperature dependence clearly. The ionic conductivity of GPE@FEC achieves 0.830×10^{-3} S/cm at room temperature, which is comparable to that of the liquid-kind electrolyte (0.729×10^{-3} S/cm), and the other value is shown in Figure 8b. As the temperature increases to 85 °C, the ionic conductivity reaches 1.557×10^{-3} S/cm. In Figure 11c, meanwhile, the ionic conductivity of both GPE@FEC and LE increases with the increase of temperature. It is noteworthy that the increase of IC in GPE@FEC is larger than that in LE, and it is significant that the difference between IC of GPE@FEC and IC of LE becomes much nearer.

The transference number of Li^+ ions is estimated by chronoamperometry and demonstrated in Figure 10e. The transference number of Li^+ (t_{Li^+}) is another vital factor for the application of all electrolytes. As mentioned above, one of the major challenges for LE, as well as some solid-state electrolytes, is the low t_{Li^+} and severe polarization due to the dual ion conduction. After CA and EIS tests, the transference number of Li^+ can be calculated by the formula 1. Therefore, the t_{Li^+} in the GPE@FEC system is 0.674, which is relatively high among the reported solid or quasi-solid electrolytes (about 0.5–0.6) and the liquid electrolyte (about 0.2–0.3). The high value of t_{Li^+} is consistent with our conclusions shown above. The strong electron-withdrawing capability of polar functional groups like $-\text{NH}_2$, $-\text{NH}-$, $\text{C}-\text{N}-\text{C}$, $-\text{OH}$, and $\text{C}-\text{O}-\text{C}$ plays an important role in this result.

This gel polymer electrolyte shows a better electrochemical performance in terms of rate, cycling and the transference number of Li^+ , as well as better ionic conductivity than the others organized and listed in Table S1.

4. CONCLUSIONS

In summary, we proposed a polymer network consisting of PEI- and PEGDE-based GPE electrolytes with the addition of FEC via a facile and reliable in situ thermal polymerization. The GPE@FEC electrolyte possessed an extremely high transference number of Li^+ ($t_{\text{Li}^+} = 0.674$) and high ionic conductivity ($0.830 \times 10^{-3} \text{ S/cm}$ at 25°C and $1.557 \times 10^{-3} \text{ S/cm}$ at 85°C), which was comparable to the ionic conductivity of commercial LE. The gel electrolyte precursor could easily penetrate the battery, and the cross-linking addition polymerization occurred without any initiators, which ensured strong interfacial adhesion between electrodes and the separator. In addition, the presence of FEC could facilitate the polymerization and promote the formation of a smooth SEI. These results demonstrated that this reliable GPE electrolyte may be one kind of promising electrolyte for the application of rechargeable quasi-solid-state Li–S batteries.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.1c16148>.

Diagrammatic sketch of preparation of precursors; thickness of the PE separator; ^1H NMR spectra of PEGDE, PEI, and Li–S GPE; XPS spectra of different elements in the PE separator; SEM mapping of the S element on the GPE membrane and the PE separator; and thermal properties of GPE membrane (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Xiaodong Guo – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China; orcid.org/0000-0003-0376-7760; Email: xiaodong2009@163.com

Yanxiao Chen – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China; Email: yxchen888@163.com

Authors

Tongwei Zhang – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Jun Zhang – Ruyuan Dongyangguang Magnetic Material Co., Ltd., Shaoguan 512600, P. R. China

Shan Yang – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Yuan Li – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Ran Dong – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Jialiang Yuan – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Yuxia Liu – The Key Laboratory of Life-Organic Analysis, Key Laboratory of Pharmaceutical Intermediates and Analysis of National Medicine, School of Chemistry and Chemical Engineering, Qufu Normal University, Qufu 273165, P. R. China; orcid.org/0000-0003-1139-8563

Zhenguo Wu – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China; orcid.org/0000-0002-8153-2169

Yang Song – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Yanjuan Zhong – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China; orcid.org/0000-0002-5057-3144

Wei Xiang – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China; orcid.org/0000-0002-8667-5257

Benhe Zhong – School of Chemical Engineering, Sichuan University, Chengdu 610065, P. R. China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acsami.1c16148>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (Grant Nos. 21878195, Nos. 21805198 and U20A20145), The National Key R&D Program of China (No. 2019YFC1906402), the Distinguished Young Foundation of Sichuan Province (No. 2020JDJQ0027), the Key Research and Development Program of Sichuan Province, China (No. 2020YFG0471 and No. 2020YFG0022), Sichuan Province Science and Technology Achievement Transfer and Transformation Project (No. 21ZHSF0111), Research Foundation for the Sichuan University and Zigong City Joint research project, China (No. 2018CDZG-16), the Distinguished Young Foundation, of the state key Laboratory of Polymer Materials Engineering, Polymer Research Institute of Sichuan University, China, (SKLPME2020-3-02) and Sichuan University postdoctoral interdisciplinary Innovation Fund (No. 2021SCU12084).

■ REFERENCES

- (1) Chen, C.; Yu, T.; Yang, M.; Zhao, X.; Shen, X. An All-Solid-State Rechargeable Chloride Ion Battery. *Adv. Sci.* **2019**, *6*, No. 1802130.
- (2) Zhou, Q.; Ma, J.; Dong, S.; Li, X.; Cui, G. Intermolecular Chemistry in Solid Polymer Electrolytes for High-Energy-Density Lithium Batteries. *Adv. Mater.* **2019**, *31*, No. 1902029.
- (3) Song, Y.; Yang, L.; Zhao, W.; Wang, Z.; Zhao, Y.; Wang, Z.; Zhao, Q.; Liu, H.; Pan, F. Revealing the Short-Circuiting Mechanism of Garnet-Based Solid-State Electrolyte. *Adv. Energy Mater.* **2019**, *9*, No. 1900671.

- (4) Kalluri, S.; Yoon, M.; Jo, M.; Liu, H. K.; Dou, S. X.; Cho, J.; Guo, Z. Feasibility of Cathode Surface Coating Technology for High-Energy Lithium-ion and Beyond-Lithium-ion Batteries. *Adv. Mater.* **2017**, *29*, No. 1605807.
- (5) Li, Q.; Liu, Y.; Wang, Y.; Chen, Y.; Guo, X.; Wu, Z.; Zhong, B. Review of the Application of Biomass-derived Porous Carbon in Lithium-sulfur Batteries. *Ionics* **2020**, *26*, 4765–4781.
- (6) Yin, Y.-X.; Xin, S.; Guo, Y.-G.; Wan, L.-J. Lithium–Sulfur Batteries: Electrochemistry, Materials, and Prospects. *Angew. Chem., Int. Ed.* **2013**, *52*, 13186–13200.
- (7) Yu, X.; Manthiram, A. Electrode–Electrolyte Interfaces in Lithium–Sulfur Batteries with Liquid or Inorganic Solid Electrolytes. *Acc. Chem. Res.* **2017**, *50*, 2653–2660.
- (8) Fang, R.; Zhao, S.; Sun, Z.; Wang, D.-W.; Cheng, H.-M.; Li, F. More Reliable Lithium–Sulfur Batteries: Status, Solutions and Prospects. *Adv. Mater.* **2017**, *29*, No. 1606823.
- (9) Cao, Y.; Zuo, P.; Lou, S.; Sun, Z.; Li, Q.; Huo, H.; Ma, Y.; Du, C.; Gao, Y.; Yin, G. A Quasi-solid-state Li–S Battery with High Energy Density, Superior Stability and Safety. *J. Mater. Chem. A* **2019**, *7*, 6533–6542.
- (10) Dai, J.; Yang, C.; Wang, C.; Pastel, G.; Hu, L. Interface Engineering for Garnet-Based Solid-State Lithium-Metal Batteries: Materials, Structures, and Characterization. *Adv. Mater.* **2018**, *30*, No. 1802068.
- (11) Tang, S.; Guo, W.; Fu, Y. Advances in Composite Polymer Electrolytes for Lithium Batteries and Beyond. *Adv. Energy Mater.* **2021**, *11*, No. 2000802.
- (12) Zhang, Y.; Tao, L.; Xie, C.; Wang, D.; Zou, Y.; Chen, R.; Wang, Y.; Jia, C.; Wang, S. Defect Engineering on Electrode Materials for Rechargeable Batteries. *Adv. Mater.* **2020**, *32*, No. 1905923.
- (13) Liang, X.; Hart, C.; Pang, Q.; Garsuch, A.; Weiss, T.; Nazar, L. F. A Highly Efficient Polysulfide Mediator for Lithium-sulfur Batteries. *Nat. Commun.* **2015**, *6*, No. 5682.
- (14) Lei, D.; Shi, K.; Ye, H.; Wan, Z.; Wang, Y.; Shen, L.; Li, B.; Yang, Q.-H.; Kang, F.; He, Y.-B. Progress and Perspective of Solid-State Lithium–Sulfur Batteries. *Adv. Funct. Mater.* **2018**, *28*, No. 1707570.
- (15) Wu, X.; Chen, K.; Yao, Z.; Hu, J.; Huang, M.; Meng, J.; Ma, S.; Wu, T.; Cui, Y.; Li, C. Metal Organic Framework Reinforced Polymer Electrolyte with High Cation Transference Number to Enable Dendrite-free Solid State Li Metal Conversion Batteries. *J. Power Sources* **2021**, *501*, No. 229946.
- (16) Cho, Y. G.; Hwang, C.; Cheong, D. S.; Kim, Y. S.; Song, H. K. Gel/Solid Polymer Electrolytes Characterized by In Situ Gelation or Polymerization for Electrochemical Energy Systems. *Adv. Mater.* **2019**, *31*, No. 1804909.
- (17) Chen, W.; Lei, T.; Qian, T.; Lv, W.; He, W.; Wu, C.; Liu, X.; Liu, J.; Chen, B.; Yan, C.; Xiong, J. A. New Hydrophilic Binder Enabling Strongly Anchoring Polysulfides for High-Performance Sulfur Electrodes in Lithium-Sulfur Battery. *Adv. Energy Mater.* **2018**, *8*, No. 1702889.
- (18) Nagao, M.; Hayashi, A.; Tatsumisago, M. Electrochemical Performance of All-Solid-State Li/S Batteries with Sulfur-Based Composite Electrodes Prepared by Mechanical Milling at High Temperature. *Energy Technol.* **2013**, *1*, 186–192.
- (19) Bok, T.; Cho, S.-J.; Choi, S.; Choi, K.-H.; Park, H.; Lee, S.-Y.; Park, S. An Effective Coupling of Nanostructured Si and Gel Polymer Electrolytes for High-performance Lithium-ion Battery Anodes. *RSC Adv.* **2016**, *6*, 6960–6966.
- (20) Zeng, F.-L.; Li, N.; Shen, Y.-Q.; Zhou, X.-Y.; Jin, Z.-Q.; Yuan, N.-Y.; Ding, J.-N.; Wang, A.-B.; Wang, W.-K.; Yang, Y.-S. Improve the Electrodeposition of Sulfur and Lithium Sulfide in Lithium-sulfur Batteries with a Comb-like Ion-conductive Organo-polysulfide Polymer Binder. *Energy Storage Mater.* **2019**, *18*, 190–198.
- (21) Li, S.; Zhang, W.; Zheng, J.; Lv, M.; Song, H.; Du, L. Inhibition of Polysulfide Shuttles in Li–S Batteries: Modified Separators and Solid-State Electrolytes. *Adv. Energy Mater.* **2021**, *11*, No. 2000779.
- (22) Otoyama, M.; Sakuda, A.; Tatsumisago, M.; Hayashi, A. Sulfide Electrolyte Suppressing Side Reactions in Composite Positive Electrodes for All-Solid-State Lithium Batteries. *ACS Appl. Mater. Interfaces* **2020**, *12*, 29228–29234.
- (23) Judez, X.; Martinez-Ibañez, M.; Santiago, A.; Armand, M.; Zhang, H.; Li, C. Quasi-solid-state Electrolytes for Lithium Sulfur Batteries: Advances and Perspectives. *J. Power Sources* **2019**, *438*, No. 226985.
- (24) Ogawa, H.; Unemoto, A.; Honma, I. Quasi-Solid-State Lithium-Sulfur Battery Using Room Temperature Ionic Liquid-Li-salt-Fumed Silica Nanoparticle Composites as Electrolytes. *Electrochemistry* **2012**, *80*, 765–767.
- (25) Aram, E.; Ehsani, M.; Khonakdar, H. A. Improvement of Ionic Conductivity and Performance of Quasi-solid-state Dye Sensitized Solar Cell Using PEO/PMMA Gel Electrolyte. *Thermochim. Acta* **2015**, *615*, 61–67.
- (26) Wang, M.; Yang, L.; Zhou, X.; Lin, Y.; Xueping, L.; Shujing, F.; Xiao, X. A New Type Quasi-solid State Electrolyte for Dye-sensitized Solar Cells. *Chin. Sci. Bull.* **2006**, *51*, 1551–1556.
- (27) Cheng, F.; Ou, Y.; Liu, G.; Zhao, L.; Dong, B.; Wang, S.; Wen, S. Novel Quasi-Solid-State Electrolytes based on Electrospun Poly(vinylidene fluoride) Fiber Membranes for Highly Efficient and Stable Dye-Sensitized Solar Cells. *Nanomaterials* **2019**, *9*, No. 9050783.
- (28) Wang, X.; Zhang, Y.; Xu, Q.; Xu, J.; Wu, B.; Gong, M.; Chu, J.; Xiong, S. A Low-cost Quasi-solid DSSC Assembled with PVDF-based Gel Electrolyte Plasticized by PC–EC & Electrodeposited Pt Counter Electrode. *J. Photochem. Photobiol., A* **2015**, *311*, 112–117.
- (29) Song, Y. X.; Shi, Y.; Wan, J.; Liu, B.; Wan, L. J.; Wen, R. Dynamic Visualization of Cathode/Electrolyte Evolution in Quasi-Solid-State Lithium Batteries. *Adv. Energy Mater.* **2020**, *10*, No. 2000465.
- (30) Hou, T.-Z.; Chen, X.; Peng, H.-J.; Huang, J.-Q.; Li, B.-Q.; Zhang, Q.; Li, B. Design Principles for Heteroatom-Doped Nanocarbon to Achieve Strong Anchoring of Polysulfides for Lithium–Sulfur Batteries. *Small* **2016**, *12*, 3283–3291.
- (31) Wang, T.; Zhai, P.; Legut, D.; Wang, L.; Liu, X.; Li, B.; Dong, C.; Fan, Y.; Gong, Y.; Zhang, Q. S-Doped Graphene-Regional Nucleation Mechanism for Dendrite-Free Lithium Metal Anodes. *Adv. Energy Mater.* **2019**, *9*, No. 1804000.
- (32) Hou, T.-Z.; Xu, W.-T.; Chen, X.; Peng, H.-J.; Huang, J.-Q.; Zhang, Q. Lithium Bond Chemistry in Lithium–Sulfur Batteries. *Angew. Chem., Int. Ed.* **2017**, *56*, 8178–8182.
- (33) Peng, H.-J.; Hou, T.-Z.; Zhang, Q.; Huang, J.-Q.; Cheng, X.-B.; Guo, M.-Q.; Yuan, Z.; He, L.-Y.; Wei, F. Strongly Coupled Interfaces between a Heterogeneous Carbon Host and a Sulfur-Containing Guest for Highly Stable Lithium-Sulfur Batteries: Mechanistic Insight into Capacity Degradation. *Adv. Mater. Interfaces* **2014**, *1*, No. 1400227.
- (34) Hou, T. Z.; Chen, X.; Peng, H. J.; Huang, J. Q.; Li, B. Q.; Zhang, Q.; Li, B. Design Principles for Heteroatom-Doped Nanocarbon to Achieve Strong Anchoring of Polysulfides for Lithium-sulfur Batteries. *Small* **2016**, *12*, 3283–91.
- (35) Zhou, Q.; Ma, J.; Dong, S.; Li, X.; Cui, G. Intermolecular Chemistry in Solid Polymer Electrolytes for High-Energy-Density Lithium Batteries. *Adv. Mater.* **2019**, *31*, No. 1902029.
- (36) Bruce, P. G.; Vincent, C. A. Polymer Electrolytes. *J. Chem. Soc., Faraday Trans.* **1993**, *89*, 3187–3203.
- (37) Hu, J.; Chen, K.; Yao, Z.; Li, C. Unlocking Solid-state Conversion Batteries Reinforced by Hierarchical Microsphere Stacked Polymer Electrolyte. *Sci. Bull.* **2021**, *66*, 694–707.
- (38) Chen, H.-W.; Jiang, C.-H.; Wu, H.-D.; Chang, F.-C. Hydrogen Bonding Effect on the Poly(ethylene oxide), Phenolic resin, and Lithium Perchlorate-based Solid-state Electrolyte. *J. Appl. Polym. Sci.* **2004**, *91*, 1207–1216.
- (39) Li, S.; Zhang, W.; Zheng, J.; Lv, M.; Song, H.; Du, L. Inhibition of Polysulfide Shuttles in Li–S Batteries: Modified Separators and Solid-State Electrolytes. *Adv. Energy Mater.* **2021**, *11*, No. 2000779.
- (40) Ngai, K. S.; Ramesh, S.; Ramesh, K.; Juan, J. C. A Review of Polymer Electrolytes: Fundamental, Approaches and Applications. *Ionics* **2016**, *22*, 1259–1279.

- (41) Osada, I.; de Vries, H.; Scrosati, B.; Passerini, S. Ionic-Liquid-Based Polymer Electrolytes for Battery Applications. *Angew. Chem., Int. Ed.* **2016**, *55*, 500–513.
- (42) Chen, W.; Lei, T.; Qian, T.; Lv, W.; He, W.; Wu, C.; Liu, X.; Liu, J.; Chen, B.; Yan, C.; Xiong, J. A. New Hydrophilic Binder Enabling Strongly Anchoring Polysulfides for High-Performance Sulfur Electrodes in Lithium-Sulfur Battery. *Adv. Energy Mater.* **2018**, *8*, No. 1702889.
- (43) Choi, S.-H.; Lee, K.-P.; Kang, H.-D. Immobilization of Lipase on a Polymeric Microsphere with an Epoxy Group Prepared by Radiation-induced Polymerization. *J. Appl. Polym. Sci.* **2003**, *88*, 1153–1161.
- (44) Wang, L.; Zhao, J.; Sun, Y.-Y.; Zhang, S. B. Characteristics of Raman Spectra for Graphene Oxide from ab Initio Simulations. *J. Chem. Phys.* **2011**, *135*, No. 184503.
- (45) Lee, Y.-W.; Shin, W.-K.; Kim, D.-W. Cycling Performance of Lithium-ion Polymer Batteries Assembled Using In-situ Chemical Cross-linking without a Free Radical Initiator. *Solid State Ionics* **2014**, *255*, 6–12.
- (46) Zhou, J.; Ji, H.; Liu, J.; Qian, T.; Yan, C. A New High Ionic Conductive Gel Polymer Electrolyte Enables Highly Stable Quasi-solid-state Lithium Sulfur Battery. *Energy Storage Mater.* **2019**, *22*, 256–264.
- (47) Zhang, W.; Wang, S.; Ji, J.; Li, Y.; Zhang, G.; Zhang, F.; Fan, X. Primary and Tertiary Amines Bifunctional Graphene Oxide for Cooperative Catalysis. *Nanoscale* **2013**, *5*, 6030–6033.
- (48) Yang, D.; Velamakanni, A.; Bozkolu, G.; Park, S.; Stoller, M.; Piner, R. D.; Stankovich, S.; Jung, I.; Field, D. A.; Ventrice, C. A.; Ruoff, R. S. Chemical Analysis of Graphene Oxide Films after Heat and Chemical Treatments by X-ray Photoelectron and Micro-Raman Spectroscopy. *Carbon* **2009**, *47*, 145–152.
- (49) Chen, C.-Y.; Peng, H.-J.; Hou, T.-Z.; Zhai, P.-Y.; Li, B.-Q.; Tang, C.; Zhu, W.; Huang, J.-Q.; Zhang, Q. A Quinonoid-Imine-Enriched Nanostructured Polymer Mediator for Lithium–Sulfur Batteries. *Adv. Mater.* **2017**, *29*, No. 1606802.
- (50) Fan, F. Y.; Carter, W. C.; Chiang, Y.-M. Mechanism and Kinetics of Li₂S Precipitation in Lithium–Sulfur Batteries. *Adv. Mater.* **2015**, *27*, 5203–5209.
- (51) Hu, J.; Tian, J.; Li, C. Nanostructured Carbon Nitride Polymer-Reinforced Electrolyte To Enable Dendrite-Suppressed Lithium Metal Batteries. *ACS Appl. Mater. Interfaces* **2017**, *9*, 11615–11625.
- (52) Li, W.; Pang, Y.; Zhu, T.; Wang, Y.; Xia, Y. A. Gel Polymer Electrolyte based Lithium-sulfur Battery with Low Self-discharge. *Solid State Ionics* **2018**, *318*, 82–87.
- (53) Agostini, M.; Lim, D. H.; Sadd, M.; Fasciani, C.; Navarra, M. A.; Panero, S.; Brutti, S.; Matic, A.; Scrosati, B. Stabilizing the Performance of High-Capacity Sulfur Composite Electrodes by a New Gel Polymer Electrolyte Configuration. *ChemSusChem* **2017**, *10*, 3490–3496.
- (54) Liu, M.; Zhou, D.; He, Y.-B.; Fu, Y.; Qin, X.; Miao, C.; Du, H.; Li, B.; Yang, Q.-H.; Lin, Z.; Zhao, T. S.; Kang, F. Novel Gel Polymer Electrolyte for High-performance Lithium–sulfur Batteries. *Nano Energy* **2016**, *22*, 278–289.
- (55) Fang, R.; Xu, H.; Xu, B.; Li, X.; Li, Y.; Goodenough, J. B. Reaction Mechanism Optimization of Solid-State Li–S Batteries with a PEO-Based Electrolyte. *Adv. Funct. Mater.* **2021**, *31*, No. 2001812.
- (56) Liang, J.; Sun, Q.; Zhao, Y.; Sun, Y.; Wang, C.; Li, W.; Li, M.; Wang, D.; Li, X.; Liu, Y.; Adair, K.; Li, R.; Zhang, L.; Yang, R.; Lu, S.; Huang, H.; Sun, X. Stabilization of All-solid-state Li–S Batteries with a Polymer–ceramic Sandwich Electrolyte by Atomic Layer Deposition. *J. Mater. Chem. A* **2018**, *6*, 23712–23719.
- (57) Tao, X.; Liu, Y.; Liu, W.; Zhou, G.; Zhao, J.; Lin, D.; Zu, C.; Sheng, O.; Zhang, W.; Lee, H.-W.; Cui, Y. Solid-State Lithium–Sulfur Batteries Operated at 37 °C with Composites of Nanostructured Li₇La₃Zr₂O₁₂/Carbon Foam and Polymer. *Nano Lett.* **2017**, *17*, 2967–2972.
- (58) Li, C.; Guo, Z.; Yang, B.; Liu, Y.; Wang, Y.; Xia, Y. A Rechargeable Li–CO₂ Battery with a Gel Polymer Electrolyte. *Angew. Chem., Int. Ed.* **2017**, *56*, 9126–9130.